Database Systems viva (ER modelling): answer key (examiner only)

DS1030 | do not give this page to the students

Each student is built to land in a different place, so the comparison between the examiner and the system has a real spread. The expectations below are what each script is designed to produce. The actual results will differ, and any difference is exactly what the validation is looking for.

Rubric concepts (15): Entity, Entity Set, Attribute, Composite Attribute, Multivalued Attribute, Derived Attribute, Relationship, Cardinality, One-to-One Relationship, One-to-Many Relationship, Many-to-Many Relationship, Primary Key, Foreign Key, Weak Entity, ER Modelling.

Overview

No.	Style	Words / time	Concepts mentioned	Weight covered
01	Complete and fluent	207 words, about 1:14	15 of 15	100%
02	Strong content, but rushed	218 words, about 1:04	14 of 15	94%
03	Hesitant and uncertain	224 words, about 1:25	9 of 15	66%
04	Entities and attributes only	242 words, about 1:22	6 of 15	37%
05	Confident but wrong	262 words, about 1:20	10 of 15	68%
06	Slow, quiet and trailing off	217 words, about 1:21	5 of 15	40%
07	Explains with an example	251 words, about 1:25	12 of 15	81%
08	Very short and vague	263 words, about 1:27	3 of 15	24%
09	Reads a flat list	255 words, about 1:27	8 of 15	60%

Weight covered counts every concept the student mentions, by its rubric weight (total weight 62). Mentioning is not the same as being correct: see student 05.

Student 01: Complete and fluent

Item	Detail
Length	207 words, about 1:14
Concepts mentioned	✓ Entity ✓ Entity Set ✓ Attribute ✓ Composite Attribute ✓ Multivalued Attribute ✓ Derived Attribute ✓ Relationship ✓ Cardinality ✓ One-to-One Relationship ✓ One-to-Many Relationship ✓ Many-to-Many Relationship ✓ Primary Key ✓ Foreign Key ✓ Weak Entity ✓ ER Modelling
Wrong facts	None.
What the system should show	Highest scores: strong Engagement, Audio and Transcript, and a technical score near the top.
What this tests	The top of the scale. Checks that a good answer really scores high.

Student 02: Strong content, but rushed

Item	Detail
Length	218 words, about 1:04
Concepts mentioned	✓ Entity ✓ Entity Set ✓ Attribute ✓ Composite Attribute ✓ Multivalued Attribute ✓ Derived Attribute ✓ Relationship ✓ Cardinality ✗ One-to-One Relationship ✓ One-to-Many Relationship ✓ Many-to-Many Relationship ✓ Primary Key ✓ Foreign Key ✓ Weak Entity ✓ ER Modelling
Wrong facts	None.
What the system should show	Content scores high. Speech rate lands in the "too fast" band, which lowers the Transcript score. Voice measures may suffer.
What this tests	Whether the system separates delivery problems from content knowledge.

Student 03: Hesitant and uncertain

Item	Detail
Length	224 words, about 1:25
Concepts mentioned	✓ Entity ✓ Entity Set ✓ Attribute ✗ Composite Attribute ✗ Multivalued Attribute ✗ Derived Attribute ✓ Relationship ✓ Cardinality ✗ One-to-One Relationship ✓ One-to-Many Relationship ✓ Many-to-Many Relationship ✓ Primary Key ✓ Foreign Key ✗ Weak Entity ✗ ER Modelling
Wrong facts	None, but the answers are thin.
What the system should show	Transcript score drops (hedges and fillers). Engagement and Audio lower. Technical is medium-high because the content is right.
What this tests	The hedge and filler counts, and whether tone shows in the voice score.

Student 04: Entities and attributes only

Item	Detail
Length	242 words, about 1:22
Concepts mentioned	✓ Entity ✓ Entity Set ✓ Attribute ✓ Composite Attribute ✓ Multivalued Attribute ✓ Derived Attribute ✗ Relationship ✗ Cardinality ✗ One-to-One Relationship ✗ One-to-Many Relationship ✗ Many-to-Many Relationship ✗ Primary Key ✗ Foreign Key ✗ Weak Entity ✗ ER Modelling
Wrong facts	None. It simply skips most of the question.
What the system should show	Delivery is decent. The technical score is lower because 9 of the 15 concepts are never mentioned.
What this tests	Concept coverage: unmentioned concepts must score 0, and the answer should not be held (coverage is well above 5%).

Student 05: Confident but wrong

Item	Detail
Length	262 words, about 1:20
Concepts mentioned	✓ Entity ✗ Entity Set ✓ Attribute ✓ Composite Attribute ✓ Multivalued Attribute ✗ Derived Attribute ✓ Relationship ✓ Cardinality ✓ One-to-One Relationship ✗ One-to-Many Relationship ✓ Many-to-Many Relationship ✓ Primary Key ✗ Foreign Key ✓ Weak Entity ✗ ER Modelling
Wrong facts	Every definition is wrong: entity, attribute, composite, multivalued, relationship, cardinality, one-to-one, many-to-many, primary key and weak entity. Nothing is correct.
What the system should show	Delivery scores well (confident, clear). The technical score should be very low because the facts are wrong, even though 10 concepts are "mentioned".
What this tests	The key test: does the AI judge correctness, or just mentions? A mention is not a correct answer.

Student 06: Slow, quiet and trailing off

Item	Detail
Length	217 words, about 1:21
Concepts mentioned	✓ Entity ✗ Entity Set ✓ Attribute ✗ Composite Attribute ✗ Multivalued Attribute ✗ Derived Attribute ✓ Relationship ✗ Cardinality ✗ One-to-One Relationship ✗ One-to-Many Relationship ✗ Many-to-Many Relationship ✓ Primary Key ✓ Foreign Key ✗ Weak Entity ✗ ER Modelling
Wrong facts	None, but half-finished.
What the system should show	Pace falls in the "too slow" band, sentence completion is low. Voice score and Transcript score low. The technical score is medium-low.
What this tests	Slow pace, incomplete sentences and quiet audio.

Student 07: Explains with an example

Item	Detail
Length	251 words, about 1:25
Concepts mentioned	✓ Entity ✗ Entity Set ✓ Attribute ✓ Composite Attribute ✓ Multivalued Attribute ✓ Derived Attribute ✓ Relationship ✓ Cardinality ✗ One-to-One Relationship ✓ One-to-Many Relationship ✓ Many-to-Many Relationship ✓ Primary Key ✓ Foreign Key ✓ Weak Entity ✗ ER Modelling
Wrong facts	None.
What the system should show	High Engagement and a good voice. Technical high, with entity set, one-to-one and the term ER modelling missing.
What this tests	Whether an answer given through an example is still credited as covering the concepts.

Student 08: Very short and vague

Item	Detail
Length	263 words, about 1:27
Concepts mentioned	✓ Entity ✗ Entity Set ✓ Attribute ✗ Composite Attribute ✗ Multivalued Attribute ✗ Derived Attribute ✗ Relationship ✗ Cardinality ✗ One-to-One Relationship ✗ One-to-Many Relationship ✗ Many-to-Many Relationship ✓ Primary Key ✗ Foreign Key ✗ Weak Entity ✗ ER Modelling
Wrong facts	None.
What the system should show	Low technical score (3 of 15 concepts, thinly). It should not be held: about 24% of the concept weight is covered.
What this tests	A very weak but on-topic answer: low mark, but not held.

Student 09: Reads a flat list

Item	Detail
Length	255 words, about 1:27
Concepts mentioned	✓ Entity ✓ Entity Set ✓ Attribute ✗ Composite Attribute ✗ Multivalued Attribute ✗ Derived Attribute ✓ Relationship ✓ Cardinality ✗ One-to-One Relationship ✗ One-to-Many Relationship ✗ Many-to-Many Relationship ✓ Primary Key ✓ Foreign Key ✓ Weak Entity ✗ ER Modelling
Wrong facts	None, but every answer is a one-line definition with nothing behind it.
What the system should show	Delivery is flat. The technical score is medium-low: 8 concepts are mentioned, with very little explanation, and the attribute kinds, the three cardinality types and ER modelling are missing.
What this tests	Shallow versus deep: does the system tell a list of labels from a real explanation?

What to look for across the 9 results

- Student 05 should have a LOW technical score despite fluent delivery. If it scores high, the system is crediting mentions instead of correct answers.
- Students 02, 03 and 06 should lose marks on delivery (pace, hedges and fillers, long pauses) while their content stays reasonable.
- Students 01 and 07 should score highest overall. Student 07 shows that an example-led answer is still credited.
- Students 04, 08 and 09 should sit below the top group on the technical score, for different reasons: missing concepts, vagueness, and shallow labels.
- The examiner should mark all nine without seeing the system results, using the Mark Guide.